

Challenge #2

Step 1 - Download the Dataset

I have downloaded the data set 'the [customer personality analysis](#) dataset.

This dataset is marketing campaign data consisting of 29 features (columns). These 29 features can be grouped into four types of information: People, Products, Promotion, and Place.

People: is a feature or column that contains data about customers (ID, Year_Birth, Education, Income, Kidhome, Teenhome, Dt_Customer, Recency, and Complain).

Products: which is a feature or column that contains data about the type of product purchased by the customer (MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds)

Promotion: which is a feature or column that contains data about the effectiveness of the campaign/promotion (NumDealsPurchases, AcceptedCmp1, AcceptedCmp2, AcceptedCmp3, AcceptedCmp4, AcceptedCmp5 and Response)

Type: is a feature or column that contains data about the type of customer purchase (NumWebPurchases, NumCatalogPurchases, NumStorePurchases and NumWebVisitsMonth).

Step 2 - Prepare Microscope of work

- Business Objective:

The scenario we get is to increase the revenue of a client named Vino Vineyard in wine sales.

From this dataset, one of the objectives that we can formulate is to determine better market segmentation by seeing if there is a special behaviour of customers of a certain age, a certain level of education or marital status that affects the purchasing pattern of certain products so that we can determine the target and more appropriate campaign methods to increase the number of product sales. Then, campaign data will also determine our decision to conduct campaigns for customers to reduce campaign costs.

- The data analysis SMART goal:

- Specific: We want to increase the number of products by finding a suitable market segmentation. We will look at this segmentation by analysing some of the features that may influence customer buying patterns, such as age, education level and marital status.
- Measurable: We will know this relationship by looking at blockchart visualisation and also the coefficient correlation visualisation of the features that we consider to influence product purchase.
- Achievable: This approach is considered the most realistic approach based on statistical theory known as 'Feature Engineering'.
- Relevant: This goal is very important because we will be able to recommend the target market and what type of campaign we should do because the type of campaign is highly correlated with the target market that we will target in the market.
- Time-bound: We will do this analysis using python or excel and finish within 3-7 days.

- A description of the work that will be completed:

- Some features that are considered to achieve the objectives that have been listed as: set are: Year_Birth, Education, Marital Status, Dt_Customer, MntWines, NumWebPurchases, NumCatalogPurchases, NumStorePurchases and NumWebVisitsMonth.

- Ignore features that we do not use in the analysis process to make it easier for us.
 - Change the format of some features, such as Year_Birth and Dt_Customer, to get the customer's age and to find out how long ago the customer made the last purchase transaction.
 - Calculate the distribution of the education level and marital status of the customer.
 - If there are some feature values we can group to simplify the data, then do so. Marital_Status which in the dataset has 8 classes (Married, Together, Single, Divorced, Widow, Alone, Absurd, YOLO) can be simplified into 2 (Married and Single).
 - Create a new feature by combining multiple features/columns whose information overlaps or can be combined. For example ('NumWebPurchases', 'NumCatalogPurchases', 'NumStorePurchases', 'NumDealsPurchases') can be combined into one feature/column.
 - Cleaning data that has duplicates or missing values.
 - Visualise the feature we want to see: the correlation coefficient, which shows the relationship between two or more variables.
- **A summary of how the existing data might help develop the future strategy.**
This data set can help the Company determine marketing campaign strategies based on previous historical data.
For example, the strategy to determine the target customer in marketing (market segment) for wine marketing can be based on the level of education or income and the number of members in the family.
 - **Any early-identified potential biases/risks/issues present in this dataset that could affect the analysis.**
This includes the amount of missing data in the data set, which will result in less accurate analysis results. Some duplicated data will also impact the analysis results.
 - **Additional data that may need to be sourced (if applicable)**
We need to get data about the varieties of wines in the market (from green grapes or red grapes), the brewing styles, the percentage of alcohol in wines (0%, 10% or any),
 - **Next step**
For further analysis, we can cluster this dataset. Of course, with a clustering method such as K-means or other methods.

Step 3 - Share your work & give feedback (optional)